

Processando config 12, seed 4, teste: Alphabit
xx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xx

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed4.bin

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	2.87
p-value of test	:	0.24

CPU time used : 00:00:00.01

Generator state:
16777216 bits have been read.

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smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 12.36
p-value of test : 0.14

CPU time used : 00:00:00.00

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 117.22
p-value of test : 0.74

CPU time used : 00:00:00.00

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed4.bin

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32558.48
p-value of test : 0.79

CPU time used : 00:00:00.02

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 150.97
p-value of test : 0.92

CPU time used : 00:00:00.00

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed4.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 307.40
p-value of test : 0.45

CPU time used : 00:00:00.00

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed4.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 1.91
p-value of test : 0.03

CPU time used : 00:00:00.00

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config12_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 37.02
p-value of test : 0.25

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 24.99
p-value of test : 0.77

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 37.92
p-value of test : 0.22

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 11.77
p-value of test : 0.99

Test on the values of the Statistic C

Number of degrees of freedom : 15
ChiSquare statistic : 10.80
p-value of test : 0.77

CPU time used : 00:00:00.06

Generator state:
134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom : 62

ChiSquare statistic : 67.13
p-value of test : 0.31

Test on the values of the Statistic M

Number of degrees of freedom : 62
ChiSquare statistic : 60.43
p-value of test : 0.53

Test on the values of the Statistic J

Number of degrees of freedom : 160
ChiSquare statistic : 170.26
p-value of test : 0.27

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 41.44
p-value of test : 0.94

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 33.32
p-value of test : 0.40

CPU time used : 00:00:00.06

Generator state:
150994688 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.22

All tests were passed